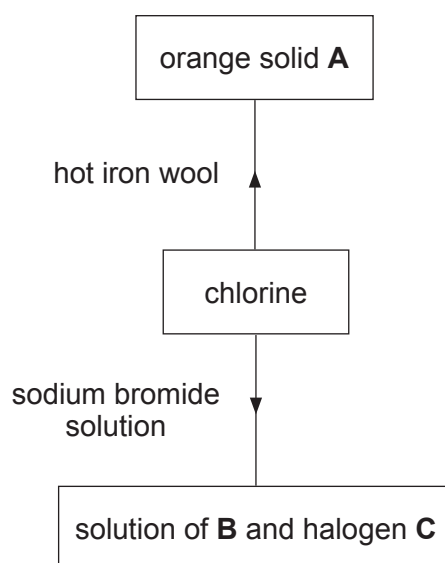


5. (a) The diagram shows some reactions of chlorine.



- (i) Give the chemical **formulae** for substances **A**, **B** and **C**. [3]

A

B

C

- (ii) State, giving a reason, the **main** safety precaution that must be taken when carrying out these reactions. [1]

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- (iii) State and explain how the observations for the reaction with sodium bromide would be different if the chlorine were replaced with iodine. [3]

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- (b) Chlorine can form a number of compounds by reacting with oxygen gas. One of these is used in water treatment.

A sample of this compound contains 0.71 g of chlorine and 0.64 g of oxygen. Calculate its simplest formula. [3]

Simplest formula

- (c) Silver nitrate solution is used to detect the presence of halide ions in solution.

- (i) The equation below represents the reaction between silver nitrate solution and potassium bromide solution.



Write the **ionic** equation for the reaction. **Include state symbols.** [2]

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- (ii) Write the balanced **symbol** equation for the reaction between calcium iodide solution and silver nitrate solution. [3]

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